HDOC Day-57 | C Programming

Problem: Read an integer array and pass it by pointer to max_subarray_sum(). Find the maximum sum of a non-empty contiguous subarray using Kadane's algorithm. For all-negative arrays, return the largest element.

C Program

```
#include <stdio.h>

void max_subarray_sum(int *arr, int size) {
    int current = *arr;
    int best = *arr;

    for (int i = 1; i < size; i++) {
        int value = *(arr + i);
        if (current + value > value)
            current = current + value;
        else
            current = value;
        if (current > best)
            best = current;
    }
    printf("Maximum subarray sum: %d\n", best);
}

int main(void) {
    int size;
    scanf("%d", &size);
    if (size <= 0) return 0;
    int arr[size];
    for (int i = 0; i < size; i++)
        scanf("%d", &arr[i]);
    max_subarray_sum(arr, size);
    return 0;
}
```

Sample Input and Output

```
Input:
9
-2 1 -3 4 -1 2 1 -5 4

Output:
Maximum subarray sum: 6
```

Explanation: Kadane's algorithm chooses whether to extend the current subarray or start a new one at each element. The best sum is updated throughout. The subarray 4, -1, 2, 1 gives 6. Initializing with the first element handles all-negative arrays.